

6 You are given that $\mathbf{M} = \begin{pmatrix} 4 & -9 \\ 1 & -2 \end{pmatrix}$.

(a) Prove that $\mathbf{M}^n = \begin{pmatrix} 1+3n & -9n \\ n & 1-3n \end{pmatrix}$ for all positive integers n . **[6]**

(b) A student thinks that this formula, when $n = 0$ and $n = -1$, gives the identity matrix and the inverse matrix $\mathbf{M}^{-1}$ respectively.

Determine whether the student is correct.

[3]